

Week 22 | Audit transfer before calling it a foundation model

A capstone that extends Week 7.4

Week 7.4 already introduces pretraining on diverse wake trajectories. Repeating a smaller source-to-target demonstration would not advance the course. This capstone instead asks students to audit why an apparent transfer benefit occurs: initialization, access to source information, representation quality, extra fitting effort or an inconsistent target-label budget.

The release uses source Re90 and three retained targets, Re100, Re105 and Re110. Those trajectories have appeared in earlier teaching and model selection. They are therefore mechanism-audit cases, not new blind research tests. There is no claim of a multi-equation foundation model or generalization to an unseen physical family.

Fix forecast horizon and count every label

Every arm begins from observed target frames 156:160, predicts 160:281 autonomously and is scored on 210:281. Varying the amount of adaptation data does not change that start or horizon. Label budgets consist of predetermined nonoverlapping ten-frame blocks before frame 150, plus the four initialization frames.

Block budget	Training frames	Validation frames	Initialization frames	Total target labels
4	30	10	4	44
8	60	20	4	84
15	120	30	4	154

Training blocks are spread across the development interval; validation blocks lie later and do not overlap them. Windows are formed inside each block, so the learner cannot create a transition across an unobserved gap. Source pretraining labels are a separate resource and remain in the accounting rather than disappearing into the word pretrained.

What each arm isolates

The scratch and pretrained causal decoders share the source-only POD basis and coefficient scaling. Their comparison isolates dynamics initialization conditional on that representation. Scratch is not source-free: it still benefits from the source-fitted basis. A matched-step decoder receives the source-plus-adaptation update ceiling entirely on target training data. A target-POD MLP provides a representation control using a basis fitted only on target training labels. DMD uses adjacent pairs inside the same allowed training blocks.

The target-POD MLP changes both representation and architecture, so its result cannot attribute all improvement to the basis alone. It is a diagnostic control: a much lower target-POD floor reveals a

representation bottleneck and motivates a follow-on same-architecture representation ablation. This qualification belongs beside the result, not only in a remote limitations section.

Compute matching has more than one meaning

Three hundred source steps plus three hundred adaptation steps equal six hundred optimizer updates. They do not equal the floating-point operations or wall time of six hundred target steps: source and target batches have different sizes, and the MLP and Transformer have different parameter counts. The record therefore reports executed updates, selected checkpoint, parameter count and measured fitting time.

$$C_{\text{deployment}} = C_{\text{adapt}} + \frac{C_{\text{source}}}{N_{\text{targets}}}$$

Amortizing source cost requires a declared number of target deployments. For one target, charge the complete source cost. For many targets, show both one-time and marginal costs. Never use amortization to hide that source data were unavailable to a supposedly information-matched source-free baseline.

The fixed update budgets make the compute-control question reproducible. A checkpoint at the ceiling remains visibly flagged. These experiments do not prove that every arm is globally optimized, and a budget extension should be selected by development criteria rather than by retained evaluation rankings.

Separate representation from adaptation

A source-trained basis may represent the target wake poorly. Projecting target truth into that basis gives a lower bound on field error for every model that uses its decoder. If this floor is 25.59% and total field error is 25.60%, the neural dynamics are not responsible for almost all of the remaining field discrepancy.

$$R = E_{\text{representation}}^2 / E_{\text{field}}^2$$

The laboratory flags representation-dominated rows when this ratio exceeds 0.95. That threshold is a teaching diagnostic, not a universal law. A plot should show field error and in-subspace error side by side, with the source-POD floor on the field panel. The target-POD control has its own floor; drawing one floor for all representations would be misleading.

The error decomposition uses squared norms. Subtracting 25.59% from 25.60% does not recover a dynamics norm. The actual in-subspace error must be measured against the projected truth or obtained from the square root of a nonnegative squared-error difference when the orthogonality assumptions hold.

A bounded transfer statement

Define relative gain as one minus pretrained in-subspace error divided by matched-control in-subspace error. Positive gain means lower error in that particular comparison; negative gain is negative transfer and remains in the table. A denominator near zero makes the ratio unstable, so absolute errors should accompany it.

$$G = 1 - E_{pretrained}/E_{control}$$

A valid conclusion names target case, target-label budget, representation and compute control. For example: on one retained target and the 84-label protocol, source initialization reduced in-subspace error relative to a declared control. That statement does not establish universal sample efficiency, full-field superiority or broad foundation-model behavior.

Aggregate means over seeds should retain their ranges. The three targets are related trajectories of the same coarse solver and geometry; they are not independent samples from all fluid mechanics. Do not turn the number of rows in a results table into an inflated count of independent scientific cases.

Reproduction as an executable claim

Stored checkpoints include the model, exact POD basis, mean and scale. Stored prediction factors reconstruct every retained transfer trajectory within a stated numerical tolerance. Their low-rank storage factors are merely a serialization device, distinct from the training POD representation. The notebook reconstructs the W22 trajectories and recomputes W22 scores directly rather than displaying a Week 21 metric under a transfer heading.

The manifest records all local scientific implementation hashes, input hashes, the protocol, platform and package versions. Text hashes normalize line endings; binary arrays retain byte hashes. A fresh-run verifier must compare numerical results with tolerances, while a checksum verifier only establishes that an artifact has not changed. These are different tests.

Capstone submission and research reading

Submit a label-accounting table, compute table, three-target comparison, field/subspace plots and a claim ledger listing supported, unsupported and unresolved statements. The four coding tasks count label unions, account for updates, detect floor-dominated errors and compute signed transfer gain. The written component should identify the next control that would most reduce uncertainty, rather than merely recommending a larger model.

Herde et al., Poseidon: Efficient Foundation Models for PDEs (2024), <https://arxiv.org/abs/2405.19101>, is a research reading for comparing scope, pretraining distributions and downstream evidence. Wu et al., Transolver (2024), <https://arxiv.org/abs/2402.02366>, supports an architectural comparison with Week 15. Neither published model is reproduced by this course experiment. Students should explain what additional datasets, controls and resource accounting would be required before making a comparable claim.

Printable laboratory worksheet

Task 1: Audit information budgets

Return the cardinality of the union of all target frames used before scoring.

List the union of training, validation and initialization frame IDs for each budget. Count overlapping frames once, but do not discard validation labels merely because no gradient was computed from them. Explain why all arms must use the same forecast start before their errors are compared.

Task 2: Match compute accounting

Count source pretraining and target adaptation steps for each learned arm, without charging source cost to a source-free arm.

Build a ledger for source updates, target updates, batch sizes, parameter counts and measured times. Show both one-time source cost and any stated amortization over targets. A scratch model using the source POD still receives source information; matching optimizer steps does not make it source-free or FLOP-matched.

Task 3: Detect a representation-limited result

Flag cases where the squared representation error explains more than 95% of squared field error. Do not subtract unsquared norms.

Compute the ratio of squared representation floor to squared total error. Compare two nearly equal field errors whose in-subspace errors differ substantially. State which control changes the representation and which changes initialization. The target-POD MLP changes two factors, so it cannot identify a pure architecture effect.

Task 4: Write a bounded comparative claim

Compute the relative reduction in in-subspace error against the matched-step control. A negative value is negative transfer and must be retained.

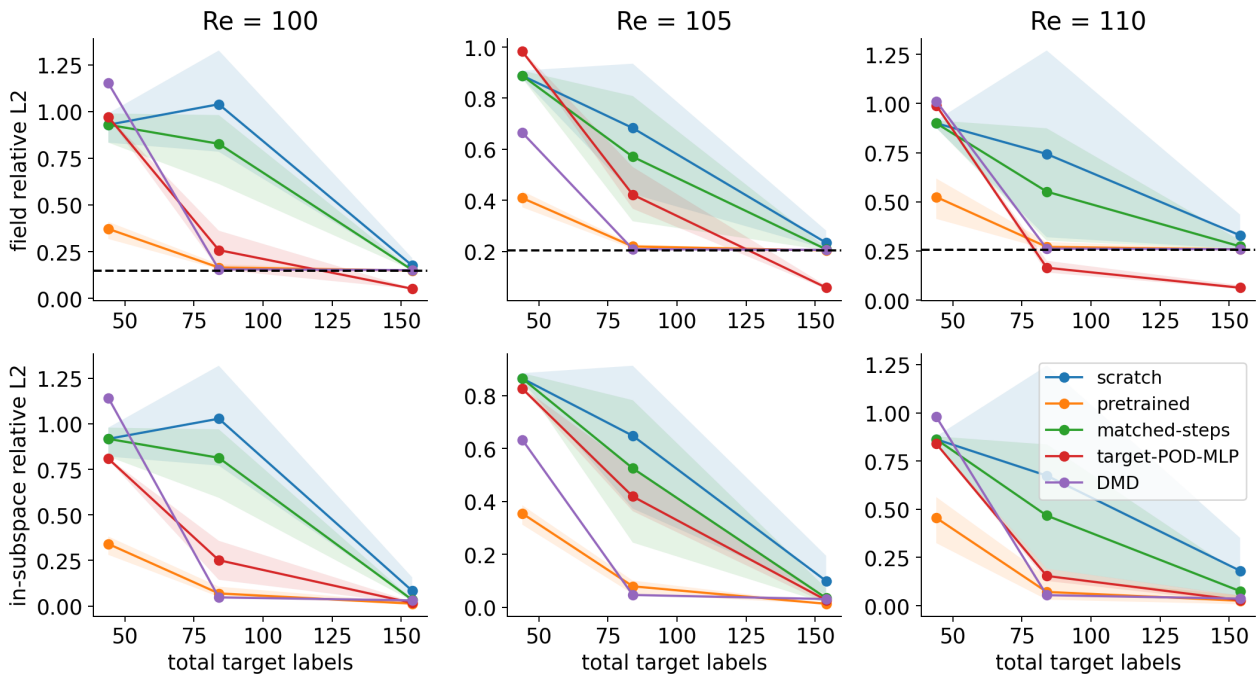
Report signed relative gain and both absolute errors for every seed. Keep negative gains and selected-step-zero outcomes. Flag a near-zero denominator instead of presenting an unstable percentage as a large benefit. End with one supported claim and one additional control needed to support a stronger claim.

Before submitting

Restart the kernel and run the worked examples and completed tasks. Confirm that all four coding checks pass and that no retained data or results file changed. Include the requested interpretation, identify every quantity fitted from training data, and separate test observations from decisions made using validation. Record unresolved budget or representation limits rather than removing inconvenient runs.

Executed evidence and interpretation

Fixed-start transfer: mean and seed range, not confidence intervals



This figure is generated from the retained protocol. Use the notebook to inspect all seeds and recompute the plotted quantities. The earlier lecture sections specify what the diagnostic can and cannot establish.